USN	1	M	S							

RAMAIAH
Institute of Technology
(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SUPPLEMENTARY SEMESTER EXAMINATIONS – JULY 2023

Program	:	**B.E :- Information Science and Engineering**	**Semester**	:	**IV**
Course Name	:	**Microcontrollers**	**Max. Marks**	:	**100**
Course Code	:	**IS46**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

1.	a)	List out the steps involved in the working of an interval timer.	CO1	(06)
	b)	List the various characteristics of an Embedded system. Also, mention few applications of embedded systems.	CO1	(06)
	c)	Discuss the four major design rules implemented on RISC design. Justify how CISC emphasizes on hardware complexity and RISC emphasizes on compiler complexity.	CO1	(08)
2.	a)	Discuss on how the ARM instruction set differs from the pure RISC definition in several ways that make the ARM instruction set suitable for embedded applications.	CO1	(06)
	b)	Discuss with a neat diagram, the Internal components of a typical MCU that works on SOC.	CO1	(08)
	c)	Briefly discuss on Pulse Width Modulator, a separate module by the virtue of its functionality that are used in most of the MCU's.	CO1	(06)

UNIT – II

3.	a)	Discuss with appropriate diagram that shows the asynchronous read and write operation for an SRAM chip type.	CO2	(08)
	b)	Discuss with appropriate diagram that illustrates the use of DMA controllers on most advanced MCUs.	CO2	(06)
	c)	Write the following numbers in the big endian and little endian format in the address 0x450011: i) 0xDC67 ii) 0x1908DEF8.	CO2	(06)
4.	a)	Discuss the various flags in embedded systems that indicate the result of each arithmetic and logical operations.	CO2	(08)
	b)	Discuss the use of memory banks for a 16- bit & 32 bit processor.	CO2	(06)
	c)	For the following operations, find out which flags will be set, (if any) for a 16-bit processor. i. Subtract 0xFE45 from 0x0978 ii. Compare A and B where A= 0x9987 and B= 0x9978.	CO2	(06)

UNIT – III

5.	a)	Discuss the various modes of operation on an ARM7TDMI processor.	CO3	(06)
	b)	Discuss on how the AMBA processor specifies the use of AHB & APB bus components.	CO3	(06)
	c)	Find out the results of the following operations given the contents of R1 = 0x0089EF32, R2 = 0x80456730 and R3=0x8 i) LSL R1, #8 ii) LSR R1, #12 iii) ASR R2, R3 iv) ROR R2, R3	CO3	(08)

Created by trial version of DocuFreezer

6.	a)	List the steps involved in the way that the processor handles an exception.	CO3	(08)
	b)	Discuss how the instruction B & BL that gives the power of decision making in an ARM program operation.	CO3	(06)
	c)	Find out the results of the following instructions. i. MOV R10,R10,LSL #2 ii. MOV R2,R2,LSR #2 iii. MOV R5,R5,ASR #2 iv. ADD R5,R5,R5,LSL #4 v. RSB R3,R4,R4,LSL #4 vi. SUB R1,R1,R1,LSL #3	CO3	(06)

UNIT – IV

7.	a)	Discuss the use of Load and Store instructions in an ARM programming with appropriate examples for each.	CO4	(10)
	b)	Write an ALP to find the average of 10 numbers stored in memory, such that the numbers are stored in Read Only memory in address starting from "TABLE".	CO4	(10)
8.	a)	Briefly discuss the internal BUS structure of the LPC 214x family with appropriate diagram.	CO4	(10)
	b)	Write an ARM C program to display the string 'I LOVE ISE' in the serial window of UART1.	CO4	(10)

UNIT-V

9.	a)	Discuss the advantages of choosing Cortex M0 by the user for design of a processor.	CO5	(08)
	b)	Discuss on three-stage pipeline for Cortex-M0 and Two-stage pipeline for cortex-M0+ with appropriate diagram for each.	CO5	(12)
10.	a)	Discuss with appropriate diagram for a memory map model for the cortex-M0 processor.	CO5	(10)
	b)	Briefly discuss the use of NVIC in the cortex processor. Also list some of its features.	CO5	(10)

****

Created by trial version of DocuFreezer